

# Patterns Without Source

*A Meta-Analysis of PWS Cup 2025*

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# We usually learn the type of method, never the detail

## In PWS Cup 2025

- Every team presents its approach at the final showcase, so the family of method is public
- Parameters, implementation and stopping rules are not — and only 2 of 24 teams released code

## And outside the contest

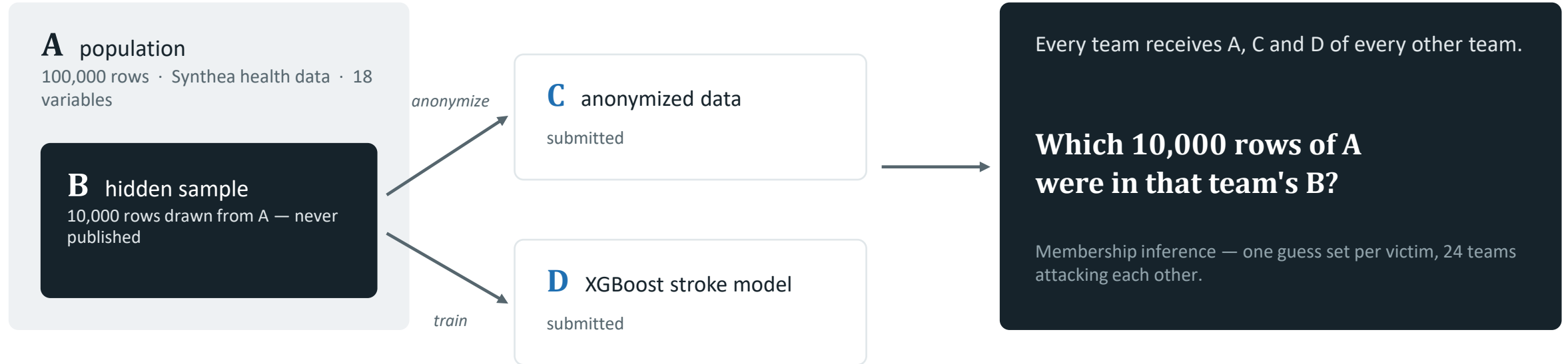
- Breach reports, news coverage and vendor statements name the type of anonymization and the type of attack
- The detail behind them is almost never published

## ? *How far can type-level descriptions alone take us?*

PWS Cup 2025 gives us 24 teams' outcomes, and a type-level description of how each of them got there.

# PWS Cup 2025: every team is both defender and attacker

## 1 Anonymization phase



**Utility score** from C and D, up to 100    **Anonymization score** =  $(10,000 - \text{best attacker's correct guesses}) / 100$ , minus a penalty

# We recovered each team's approach by hand, at the showcase

24

teams registered

20

approaches we could  
reconstruct

2

teams that released code

## 1 Attend the showcase

Posters, spotlight talks and follow-up discussions  
- the only public record of each team's approach.

## 2 Write the approaches down

Anonymization, model training and attack,  
documented team by team.

## 3 Code them as categories

Keep a feature only if at least four teams used it,  
so no effect rests on one or two teams.

*What we recovered is the family of each team's method — not its parameters, and not its code.*

# Eight categorical features — and what the two winners chose

## Defender

20 teams

**Base** B (13) → elsewhere (7)

**Synthesis** no (9) → yes (11)

**Noise** no (14) → yes (6)

**Optimization** no (14) → yes (6)

**Row swapping** no (10) → yes (10)

## Attacker

19 teams

**Attack target** anonymized data only (15) → also the released model (4)

**Attacker-trained model** no (13) → yes (6)

**Non-default distance** no (11) → yes (8)

→ marks the contrast direction: every effect size and every IRR is reported for the right-hand category.

### Ex. Team 18 — overall rank 1

elsewhere · synthesis yes · noise no · optimization yes · swapping no

Inferred Synthea's parameters with a generative model, built a pseudo-population, then picked its best 10,000 rows by simulated annealing.

### Ex. Team 8 — best attack score

anonymized data only · attacker-trained model yes · non-default distance no

A Gaussian LiRA combined with a distance attack on covariance-augmented data.

# Three questions — and how we test them with $n = 20$

## H1

Do generation- and reconstruction-based designs beat direct perturbation — on utility and on anonymization?

## H2-1

Does an anonymization design become vulnerable when the attacker also uses the released XGBoost model?

## H2-2

Does it become vulnerable when the attacker trains its own membership model on the anonymized data — LiRA, RMIA?

## Effect sizes

Hedges'  $g$  across the 20 teams, corrected for the small sample.  $|g|$  above 0.8 is a large effect.

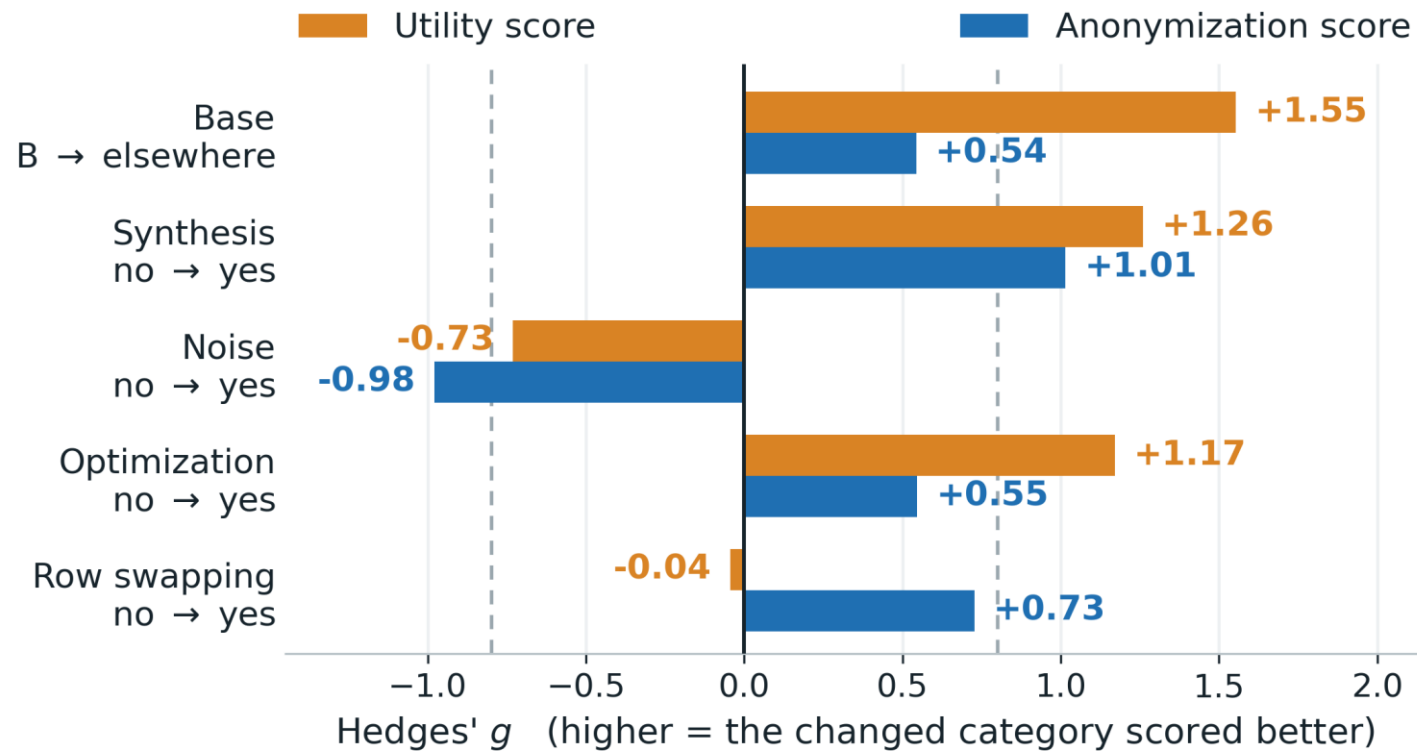
## Regression model search

A large univariate effect can come from a feature's correlation with another coded feature. The AICc search checks whether it survives with the others in the model.

## Dyadic model

Negative binomial over all 361 ordered victim-attacker pairs, so the pairing itself is in the model. Errors clustered by attacker and by victim.

# H1: reconstructing beat perturbing — on both scores



Hedges' g across 20 teams. Dashed lines mark  $|g| = 0.8$ , the conventional threshold for a large effect.

## Base (B → elsewhere)

utility +1.55

## Synthesis

utility +1.26 · anon. +1.01

## Explicit noise

utility -0.73 · anon. -0.98

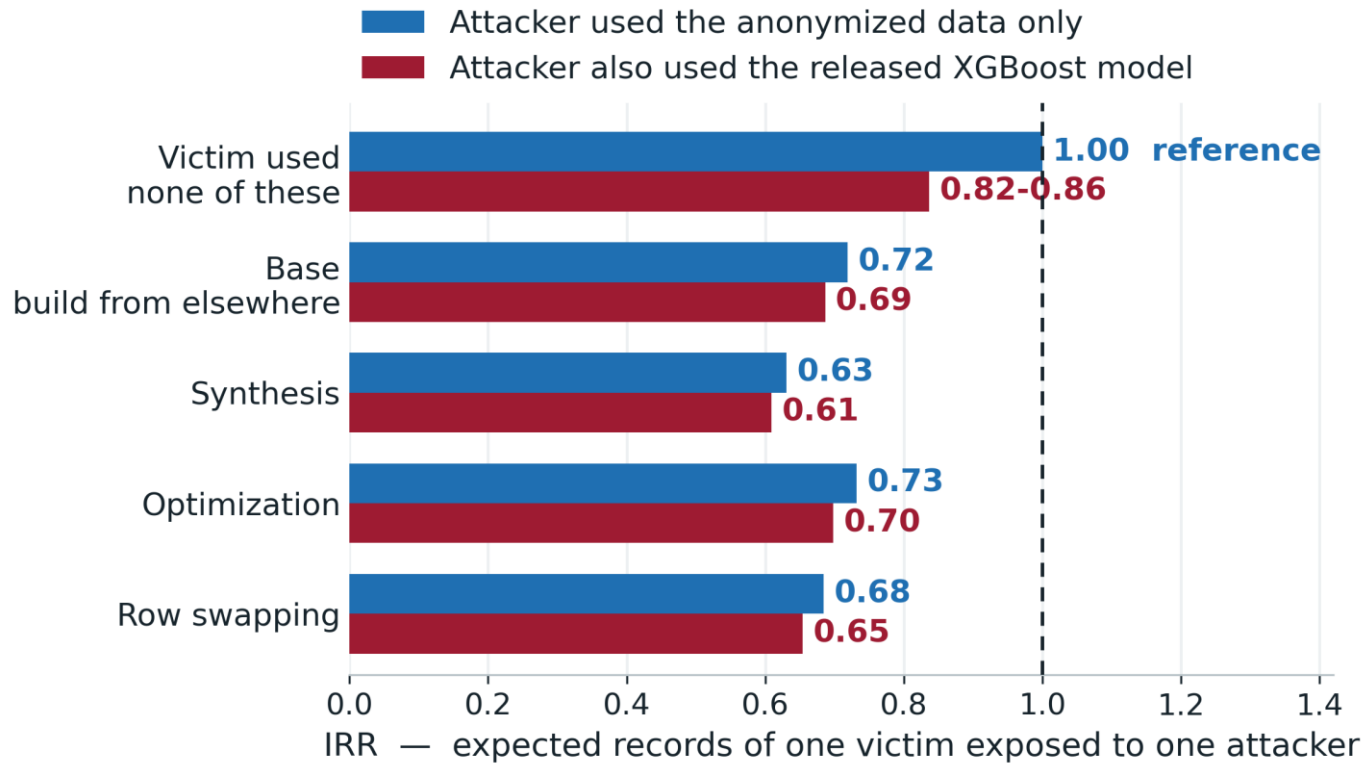
## Row swapping

utility -0.04 · anon. +0.73

The AICc search keeps Base for the statistics and Kruskal-Wallis parts, Synthesis for logistic regression.

*Synthesis is the one choice that moved both scores at once.*

## H2-1: the released model cost attackers, and only against weak targets



IRR = expected number of one victim's records exposed to one attacker; 1.00 = no design and no model use. Negative binomial, 361 ordered pairs, errors clustered by attacker and by victim.

### H2-1 partly yes

- Every design cut exposure on its own: IRR 0.63-0.73, all  $p < 0.05$
- Attackers who also used the released model landed 14-18% fewer hits — against victims using none of these designs
- Against victims that used them, the model bought nothing:  $0.72 \rightarrow 0.69$ ,  $0.63 \rightarrow 0.61$

### H2-2 no

Attacker-trained membership models — LiRA, RMIA, classifiers — showed no effect: IRR 1.04-1.06,  $p = 0.31$ -0.59, and no estimable interaction with any design. No design was specifically vulnerable to them.

# What this can and cannot support

## The constraints

- 20 teams, observed and not assigned — these are associations between design choices and scores, not causal effects
- Coded from presentations: the family of each method, not its parameters
- One contest, one Synthea population, one scoring rule

## What it still gives us

- Within PWS Cup 2025, these are the patterns the data show
- Type-level descriptions, with almost no code, were enough to produce specific and quantified hypotheses
- The same coding applies to any contest that reports approaches at that level of detail

***You do not need the code to start asking the right questions.***

# A short questionnaire would make the next contest analyzable

## What we found

### H1

Generation- and reconstruction-based designs scored higher on utility and on anonymization alike.

### H2-1

Exploiting the released model cost the attacker — and bought nothing once the victim used one of these designs.

### H2-2

Attacker-trained membership models showed no effect against any design.

## One ask for organizers

Requiring source code is not always possible — industrial and privacy constraints are real, and this contest shows what happens: two teams out of twenty-four.

**But a short questionnaire, answered by every team, costs almost nothing — and might already yield useful insight of the kind shown on the left.**

*Then the next analysis can start from what teams reported, instead of what we had to reconstruct.*

● THANK YOU

# Questions?

*Patterns Without Source: A Meta-Analysis of PWS Cup 2025*

Reconstructed data, codebook and analysis code

[github.com/kiti1924/PWSCUP2025\\_REVIEW\\_PUB](https://github.com/kiti1924/PWSCUP2025_REVIEW_PUB)

Competition materials and full results

[iwsec.org/pws/2025/cup25.html](https://iwsec.org/pws/2025/cup25.html)

Contact

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● BACKUP

# Slides for questions

# Backup: PWS Cup 2025 main results

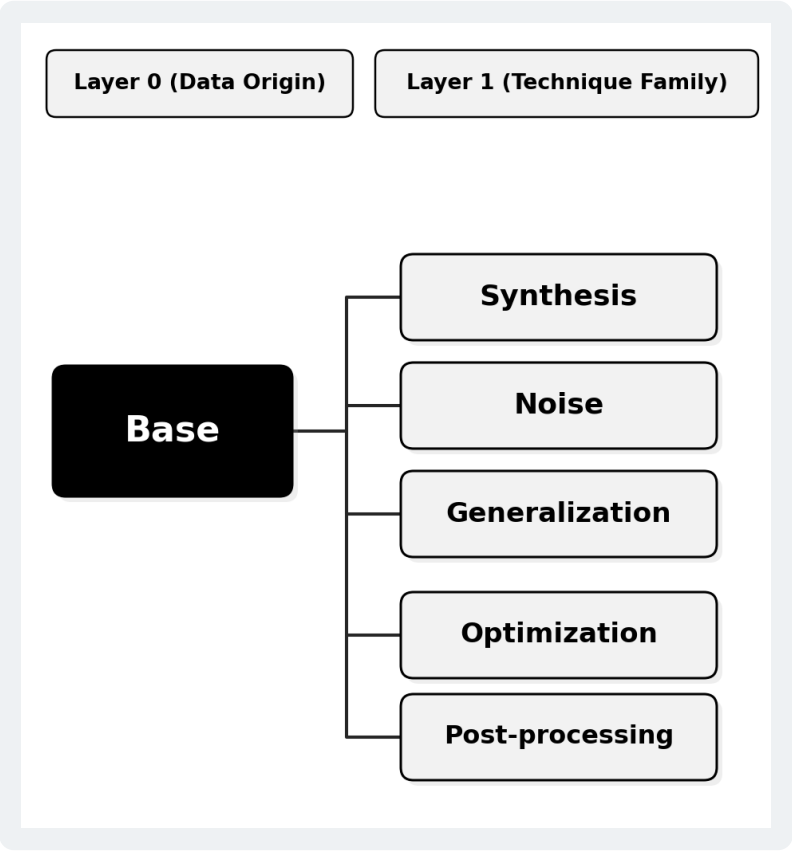
| Team | Utility | Pen. | Anon. | Total  | Rank | Attack |
|------|---------|------|-------|--------|------|--------|
| 1    | 91.89   | 13   | 70.47 | 162.36 | 8    | 113.81 |
| 2    | 91.92   | 14   | 72.86 | 164.78 | 6    | 105.43 |
| 3    | 75.93   | 20   | 65.73 | 141.66 | 17   | 105.81 |
| 4    | 77.98   | 10   | 78.60 | 156.58 | 11   | 101.47 |
| 5    | 91.64   | 19   | 66.75 | 158.39 | 10   | 107.74 |
| 6    | 91.25   | 11   | 78.31 | 169.56 | 4    | 109.17 |
| 7    | 74.09   | 17   | 71.91 | 146.00 | 16   | 103.93 |
| 8    | 91.18   | 10   | 78.63 | 169.81 | 3    | 121.13 |
| 9    | 81.04   | 13   | 74.30 | 155.34 | 12   | 106.84 |
| 10   | 91.93   | 14   | 72.21 | 164.14 | 7    | 13.79  |
| 11   | 84.54   | 14   | 69.32 | 153.86 | 13   | 104.06 |
| 12   | 78.21   | 15   | 73.90 | 152.11 | 14   | 106.60 |

| Team | Utility | Pen. | Anon. | Total  | Rank | Attack |
|------|---------|------|-------|--------|------|--------|
| 13   | 90.84   | 13   | 76.50 | 167.34 | 5    | 106.93 |
| 14   | 77.36   | 20   | 24.64 | 102.00 | 22   | 101.22 |
| 15   | 67.99   | 17   | 67.81 | 135.80 | 19   | 106.09 |
| 16   | 81.75   | 12   | 77.28 | 159.03 | 9    | 107.59 |
| 17   | 75.40   | 19   | 30.44 | 105.84 | 21   | 102.79 |
| 18   | 91.55   | 10   | 79.28 | 170.83 | 1    | 108.39 |
| 19   | 70.95   | 12   | 75.11 | 146.06 | 15   | 101.50 |
| 20   | 94.67   | 11   | 75.57 | 170.24 | 2    | 99.99  |
| 22   | 84.41   | 20   | 47.69 | 132.10 | 20   | 102.83 |
| 23   | 66.36   | 12   | 70.29 | 136.65 | 18   | 100.63 |
| 24   | 69.59   | 20   | 0.00  | 69.59  | 23   | 104.46 |

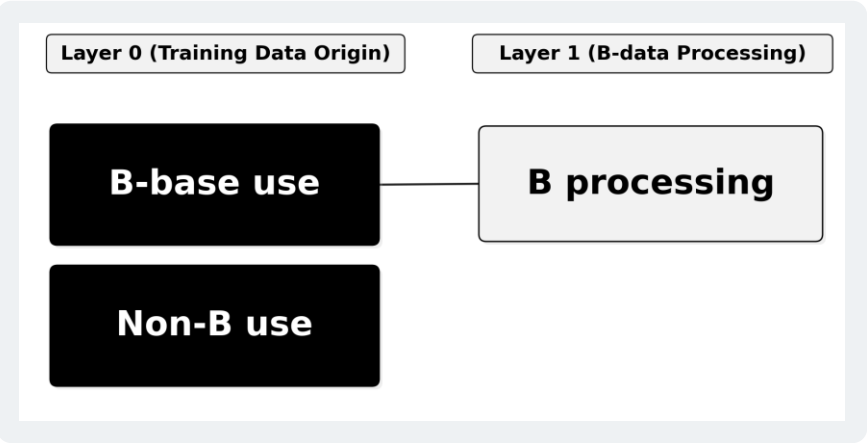
Team 21 withdrew. Team 10 submitted no attack. Final score = 0.1 x preliminary round + 0.9 x main round. The anonymization score already has the penalty subtracted.

# Backup: the full feature taxonomies

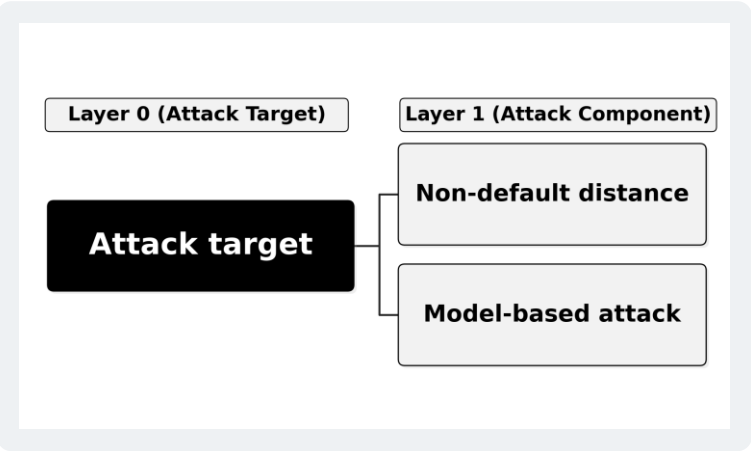
## Anonymization



## Training models

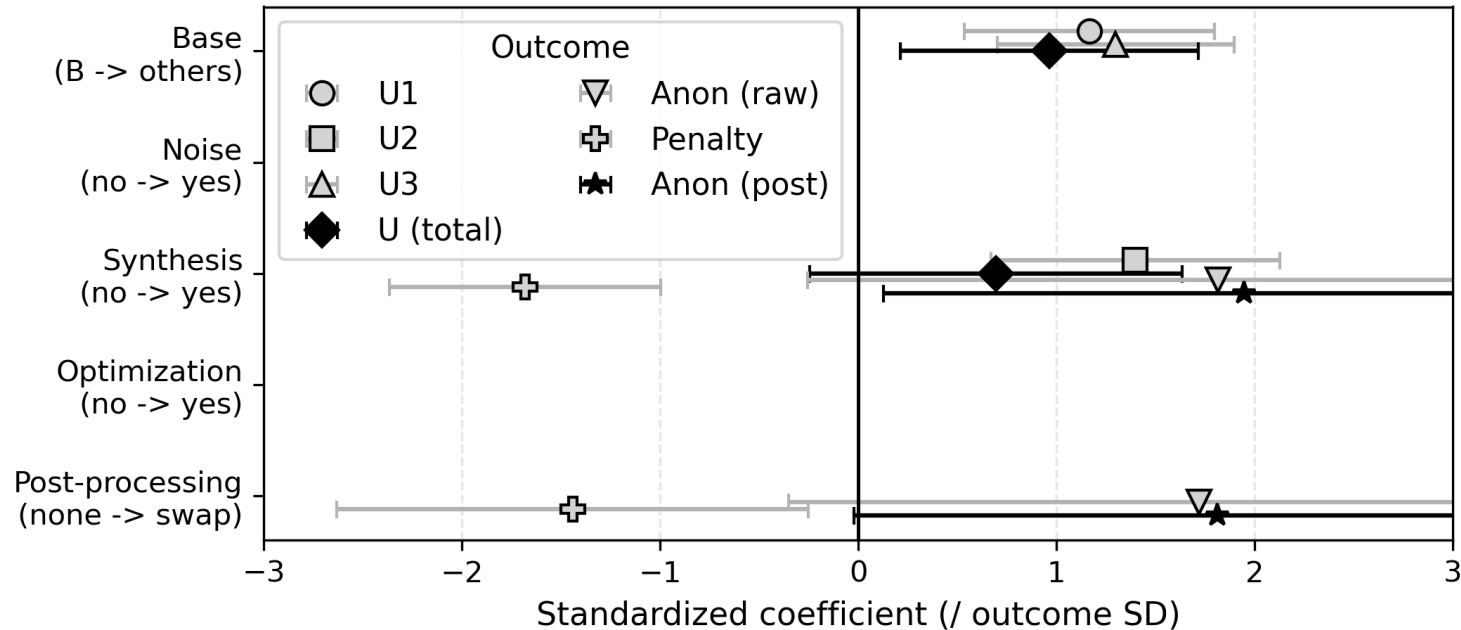


## Attacks



Features are hierarchical: a Layer 1 feature refines its Layer 0 parent. Only features present in at least four teams were kept.

# Backup: the AICc-selected regressions agree with the effect sizes



Standardized OLS coefficients, 95% CI. A missing point means the feature was not retained for that outcome.

## Two things worth saying

**Optimization had  $g = 1.17$  but was never retained.**

Four of the six teams that optimized also built from elsewhere, and optimization went with synthetic designs. Once Base or Synthesis is in the model, AICc no longer pays for it — exactly the confounding the search is there to catch.

**The anonymization models keep an interaction.**

Synthesis x swapping,  $R^2$  between 0.47 and 0.69. Swapping raises the base anonymization score by about 32 points for non-synthetic designs, and leaves synthetic ones essentially unchanged.

# Backup: team-level coding

| Team | Base   | Synth | Noise | Optim | Swap |
|------|--------|-------|-------|-------|------|
| 1    | others | Y     | N     | N     | none |
| 2    | B      | Y     | N     | N     | none |
| 3    | B      | N     | N     | N     | swap |
| 5    | others | Y     | N     | Y     | swap |
| 6    | B      | N     | Y     | N     | swap |
| 8    | others | Y     | N     | Y     | swap |
| 9    | B      | N     | Y     | N     | swap |
| 10   | others | Y     | N     | Y     | none |
| 11   | others | N     | Y     | Y     | swap |
| 12   | B      | Y     | N     | N     | none |
| 13   | B      | Y     | N     | Y     | none |
| 15   | B      | N     | Y     | N     | none |
| 16   | B      | Y     | N     | N     | swap |
| 17   | B      | N     | Y     | N     | none |
| 18   | others | Y     | N     | Y     | none |
| 19   | B      | Y     | N     | N     | swap |
| 20   | others | Y     | N     | N     | swap |
| 22   | B      | N     | N     | N     | none |
| 23   | B      | N     | N     | N     | swap |
| 24   | B      | N     | Y     | N     | none |

| Team | Attack target | Non-default dist. | Attacker-trained |
|------|---------------|-------------------|------------------|
| 1    | anon          | Y                 | N                |
| 2    | anon          | N                 | Y                |
| 3    | anon          | N                 | N                |
| 5    | anon          | Y                 | N                |
| 6    | anon          | N                 | Y                |
| 8    | anon          | N                 | Y                |
| 9    | anon          | N                 | N                |
| 10   | --            | --                | --               |
| 11   | both          | N                 | N                |
| 12   | anon          | N                 | Y                |
| 13   | anon          | Y                 | Y                |
| 15   | both          | Y                 | Y                |
| 16   | anon          | Y                 | N                |
| 17   | anon          | N                 | N                |
| 18   | anon          | Y                 | N                |
| 19   | anon          | N                 | N                |
| 20   | both          | N                 | N                |
| 22   | anon          | N                 | N                |
| 23   | anon          | Y                 | N                |
| 24   | both          | Y                 | N                |

Team 21 withdrew; teams 4, 7, 14 and 21 could not be coded. Team 10 submitted no attack.

# Backup: the full dyadic model, all eight specifications

| Victim Strategy | Baseline                       | (b) Main:<br>0.82-0.86**        | Main:<br>1.04-1.05                              |
|-----------------|--------------------------------|---------------------------------|-------------------------------------------------|
|                 | Ref<br>(1.00)                  |                                 |                                                 |
|                 | Base<br>(B -> others)          | (a) Main:<br>0.72-0.73*         | Inter: 1.03<br>(SE n.e.; Total: 0.78)           |
|                 | Synthesis<br>(no -> yes)       | Main:<br>0.63-0.65*             | Inter: 1.00<br>(SE n.e.; Total: 0.68)           |
|                 | Optimization<br>(no -> yes)    | Main:<br>0.73-0.75*             | Inter: 1.01<br>(SE n.e.; Total: 0.78)           |
|                 | Post-processing<br>(no -> yes) | Main:<br>0.68-0.70*             | Inter: 0.99<br>(SE n.e.; Total: 0.74)           |
|                 | Baseline                       | Attack target<br>(anon -> both) | Attacker-trained<br>model attack<br>(no -> yes) |
|                 | Attacker Strategy              |                                 |                                                 |

The version printed in the paper. Areas (a) victim main effects, (b) attacker main effects, (c) interactions.

## How to read the cells

### Main

The main-effect IRR, as a range across the four specifications that share that variable.

### Inter

The interaction coefficient itself — not an effect you can read on its own.

### Total

victim x attacker x interaction — the expected exposure when both are present. These are the numbers plotted on the H2-1 slide.

### SE n.e.

The two-way clustered standard error was not estimable, so no inferential claim is made — this is why H2-2 is reported as a null result rather than a small effect.

# Backup: how this sits against the synthetic-data literature

## The tension

Stadler et al. (2022) conclude that synthetic data does not give a better privacy-utility trade-off than traditional row-level sanitization.

**Here, synthesis was associated with higher scores on both axes at once.**

The two are not directly comparable. We compare across teams and whole design families, not parameters within one algorithm. Varying a parameter inside one method would still trace the classic trade-off curve.

## Other likely questions

- Differential privacy? Several teams tried DP synthesizers and abandoned them - utility was too low under this scoring rule. Only two used explicit Laplace noise.
- Why negative binomial? The exposed-record counts are heavily over-dispersed relative to Poisson.
- Why did using the released model hurt attackers? We can only speculate: the model carries little membership signal here, and the effort spent on it did not go into the distance matching that worked.